Reconstructing community dynamics from fragmented monitoring with a state-space neural ordinary differential equation

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Abstract

Tracking community dynamics is essential for assessing the resilience of ecological communities and the vital ecosystem functions and services they provide. However, although population abundance and assemblage monitoring schemes are now extensive, we lack dedicated community monitoring infrastructure, constraining our understanding of how communities have changed. Here, we introduce a modelling approach designed to harmonise data across monitoring schemes, allowing us to reconstruct community dynamics. To demonstrate this approach, we simulate ecological communities across a 50 × 50-cell landscape using a bioenergetic food-web model. We then impose sampling strategies with contrasting spatial footprints and temporal coverage. We assess how well our model—a state-space neural ordinary differential equation (SS-NODE)—can account for sampling structure and observation error to reconstruct latent community dynamics. SS-NODE accurately reconstructs biomass, richness, species pools and species interaction sensitivities through time using fragments of population time series and ancillary information on potential feeding links and phylogeny. These results demonstrate the potential to reconstruct community dynamics from existing population monitoring data.

1 Introduction

Threats such as climate change and habitat loss are putting biodiversity under intense pressure. Tracking how biodiversity responds to these threats is central to assessing the resilience of ecological communities and the vital ecosystem functions and services they provide. This need has driven the rapid expansion of biodiversity monitoring infrastructure and indicators. One common and influential approach tracks population-level change, describing whether collections of species are increasing or decreasing (Butchart et al., 2010). Although influential, these metrics provide only a partial picture of biodiversity change and do not reveal how wider communities are responding, including changes in function, stability and resilience (Dornelas et al., 2014). We therefore need metrics that describe community dynamics and identify areas at risk of community collapse.

Tracking community dynamics at large scales is constrained by a lack of community monitoring infrastructure, despite the expansion of population- and assemblage-level monitoring. Integrating these existing schemes offers an opportunity to reconstruct communities, but presents substantial challenges. Schemes sample different locations, cover different periods and observe patches of different sizes, while differences in detectability and sampling conditions further obscure the ecological signal. Reconstruction therefore requires an approach that combines these fragmented records, separates observation

error and sampling structure from the underlying ecological process, and draws information across schemes to infer community trajectories that have never been observed in full.

Existing methods address different aspects of this reconstruction problem. Joint species models combine environmental responses and residual associations to describe community composition (Warton et al., 2015), but these associations do not establish how species influence one another's subsequent dynamics (Blanchet et al., 2020). Empirical dynamic modelling provides a way to learn these relationships without specifying every underlying ecological mechanism, including interaction sensitivities that change with community state (Ye et al., 2015; Deyle et al., 2016). Applying this approach across monitoring schemes, however, requires the ecological states driving those dynamics to be inferred from monitoring schemes with different observation and sampling processes and levels of error. State-space models provide the structure for distinguishing these latent states from imperfect measurements (Auger-Méthé et al., 2021), but can be computationally demanding, limiting their application at the community scale. The methodological challenge is therefore to bring flexible community dynamics and explicit observation modelling into a common framework: one that learns how species effects vary with environment and community composition, maps latent states to each survey's spatial footprint, and remains computationally feasible across many species and locations.

Here we introduce a state-space neural ordinary differential equation (SS-NODE) for this purpose. Its observation model represents which species each scheme samples, the spatial footprint over which they are measured, and how sampling conditions affect detection. Its process model uses a neural ordinary differential equation to learn community dynamics as a shared function of species biomass, environmental conditions and species information (Chen et al., 2018). Joint fitting allows partial observations to constrain latent trajectories, while the learned dynamics connect those observations across space, species and time. We evaluate the approach using a bioenergetic food-web simulation to generate community dynamics across a landscape. We then impose three sampling designs, each representing a distinct monitoring scheme with different duration and spatial coverage. At locations excluded from learning the shared dynamics, we test whether SS-NODE reconstructs species biomass, community membership and local dynamical interaction sensitivities under two conditions: observation-assisted reconstruction, where available monitoring fragments inform local states, and ancillary-only reconstruction, where no local abundance observations are supplied. Both conditions are evaluated against identical missing-information targets.

2 Methods

We distinguish the ecological system used to generate data from the approximation fitted to monitoring records. The shared ecological approximation has a deterministic neural transition and a stochastic observation distribution; observation-assisted inference additionally permits regularised state discrepancies around that transition. This state-space structure links surveys with different spatial supports and historical coverage to one evolving community, rather than treating each scheme as a separate ecological system. The fitted model is not supplied with the bioenergetic equations below.

2.1 Ecological process

Let B_{ict} denote latent biomass of species i in cell c at time t. We simulate 18 producers and 12 consumers on a 50 by 50 landscape of 200 m cells. Consumers are assigned to monitoring assemblages of nine and three species independently of trophic position. Biomass changes through production, metabolism, environmental mortality and saturating consumption:

$$\frac{dB_i}{dt} = B_i\{b_i r_i(1 - B_i/K) - x_i - 0.36(E - \theta_i)^2\} + \sum_j e_j C_{ij} - \sum_k C_{ki}, \qquad C_{ij} = \frac{8 x_i B_i F_{ij} B_j}{0.45 + \sum_k F_{ik} B_k}. \quad (1)$$

Here b_i identifies producers, $K = 5.3125$, $r_i = 0.85\exp(-0.25\log M_i)$ and $x_i = 0.09\exp[-0.25(\log M_i - 2.4)]$. Resource assimilation efficiency is 0.65 for producers and 0.8 for consumers. Consumers have no additional self-regulation term: resource limitation and predation provide feedback. This is an allometric bioenergetic variant with a multispecies Type II functional response, motivated by body-size-dependent consumer–resource dynamics and allometric food-web models (Yodzis and Innes, 1992; Brose et al., 2006).

A simulated coalescent phylogeny defines a centred, normalised shared-ancestry covariance with a small independent component. Independent Gaussian draws under that covariance determine body-mass deviations and environmental optima, representing phylogenetically structured ecological differences (Felsenstein, 1985). Baseline log mass is zero for producers and 2.4 for consumers, with phylogenetic deviation scale 0.45; environmental optima use scale 0.8. The environmental-mismatch mortality coefficient is 0.36, yielding species-specific filtering through differences in environmental optima. Ecological integration and post-fitting assessment use double precision to retain very small positive biomasses; optimisation uses single precision. The equations are identical across precisions. Feeding preferences are Gaussian in log consumer-resource mass ratio, centred on 2 with width 0.7 and restricted to mass ratios between exp(0.6) and exp(3.4). Preferences are normalised within each consumer. Consumers may eat suitably smaller producers or consumers, permitting omnivory independently of monitoring membership. This bounded size niche is motivated by the role of body size in feeding relationships (Petchey et al., 2008); it does not implement the complete optimal-foraging Allometric Diet Breadth Model.

All spatial structure arises from the environmental field

$$E(x, y, t) = 2x/49 - 1 + 0.65\sin(2\pi y/50 + 2\pi t/24). \quad (2)$$

There is no dispersal, separate spatial random effect, feeding noise or ecological process noise. Environmental mismatch increases mortality, allowing community composition and realised interaction strengths to change despite a fixed potential feeding matrix. All biomasses start at 0.3. Eight exponential midpoint substeps integrate each year in log biomass, with environment held constant within the interval. A ten-year burn-in precedes states at years 0-30. Membership is defined as biomass greater than 0.02; lower biomasses remain in the dynamical system.

An ecology-only screen requires numerically valid trajectories (no NaN or infinity, with biomass above zero) and at least one species from each monitoring assemblage above the membership threshold in at least 80% of years. It retains 2304 cells and excludes 196 before monitoring is generated. Excluded trajectories remain saved, while monitoring and assessment are restricted to eligible cells. The screen uses no fitted predictions or recovery scores, so results are conditional on persistent multi-assemblage communities.

Figure 1 illustrates the environmental field and the resulting local dynamical sensitivities at three locations and times.

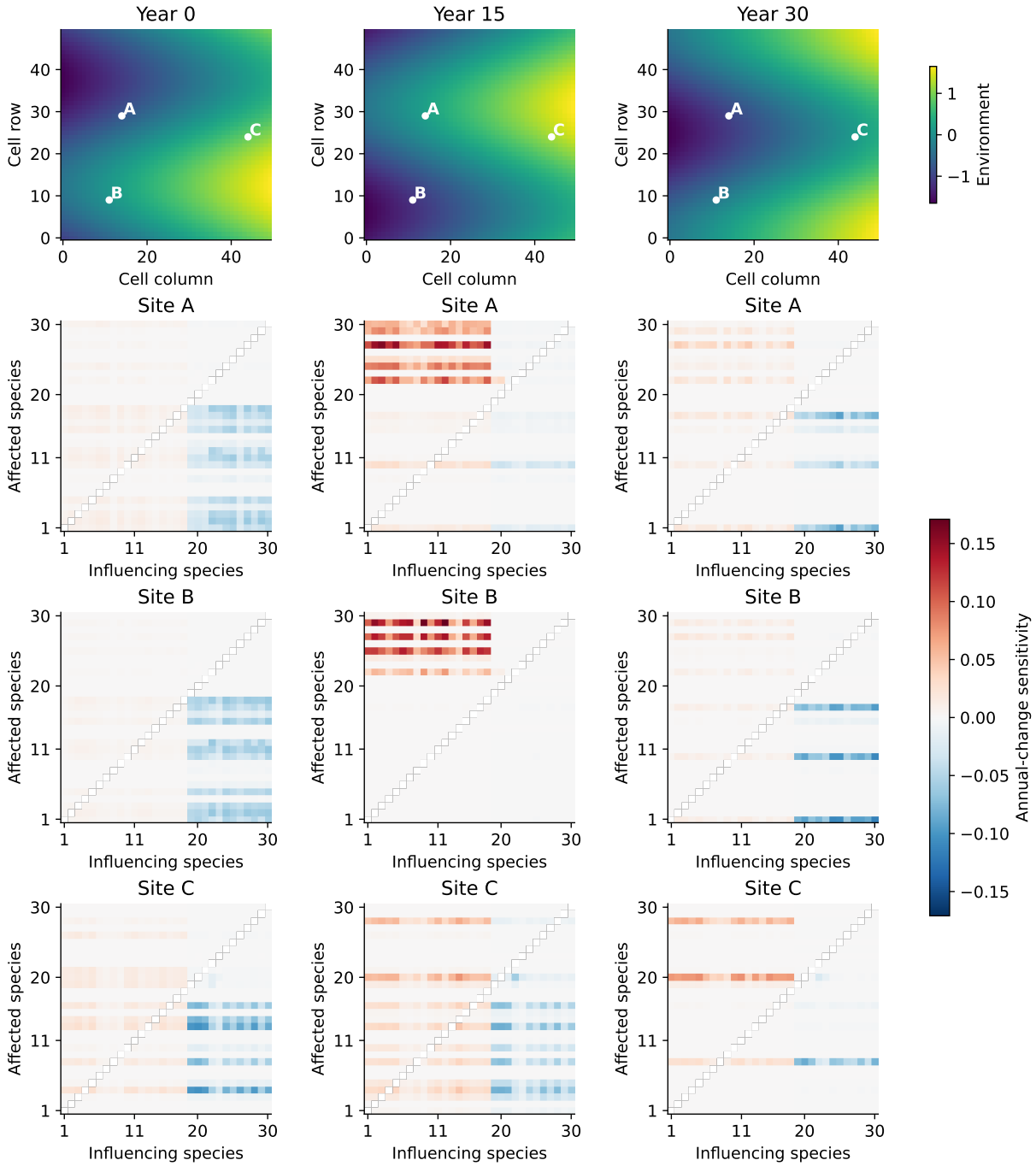


Figure 1: Environmental change and emergent local interactions. Columns show years 0, 15 and 30. Environmental maps identify the same three sites throughout; the rows below show their true off-diagonal annual-change Jacobians. Environmental and interaction colour scales are shared across the corresponding panels. The three sites were selected from initial environmental quantiles among eligible cells before assessment.

2.2 Observation process

Producer, consumer-A and consumer-B monitoring uses 1 by 1, 1 by 2 and 1 by 4 cell footprints, respectively. Each scheme targets 10% of eligible cells, rounded down to complete footprints. This gives 230, 115 and 57 patches, covering 230, 230 and 228 cells, respectively. Nominal coverage is defined relative to eligible cells; realised full-landscape coverage is 9.20%, 9.20% and 9.12%, respectively. Footprints cannot overlap within a scheme but may overlap between schemes. Disjoint horizontal strips are randomly selected within 5 by 5 tiles, with random packing offsets. This is a constrained design rather than uniform sampling over all possible rectangles. Final footprints are retained at random from a prespecified candidate survey set within each assemblage and partition. Ecological trajectories and ancillary information are generated independently of survey retention. Producer histories span years 21-30, consumer A histories years 11-30 and consumer B histories years 1-30; every scheduled year is sampled.

We distinguish latent abundance from the observation mechanism, as in hierarchical abundance models (Royle, 2004). Here counts are calibrated biomass-linked observations, rather than an N-mixture population model. A known footprint-averaging operator W maps cell biomasses to each survey support (Gotway and Young, 2002):

$$Y_{kit} \sim \text{Poisson}\left(1000 q_{kt} \sum_{c} W_{kc} B_{cit}\right), \qquad \text{logit}(q_{kt}) = \text{logit}(0.7) + \beta_{g(k)} Q_{kt}. \tag{3}$$

Survey quality Q is recorded and uniform on [-1,1]. One coefficient per scheme is drawn from Normal(0.7,0.2), shared across its species, hidden from fitting and estimated jointly with the dynamics. Poisson variation is the only random observation noise; aggregation itself is deterministic. Effort and biomass-to-count conversion are fixed at 1,000, and larger footprints do not automatically receive greater effort. Counts represent calibrated biomass-linked abundance. Environmental histories are available at constituent-cell resolution, allowing the model to reconstruct cells before averaging. Missing species or historical records are masked, never interpreted as zeros.

Figure 2 summarises the spatial footprints and unequal monitoring histories supplied to the fitted model.

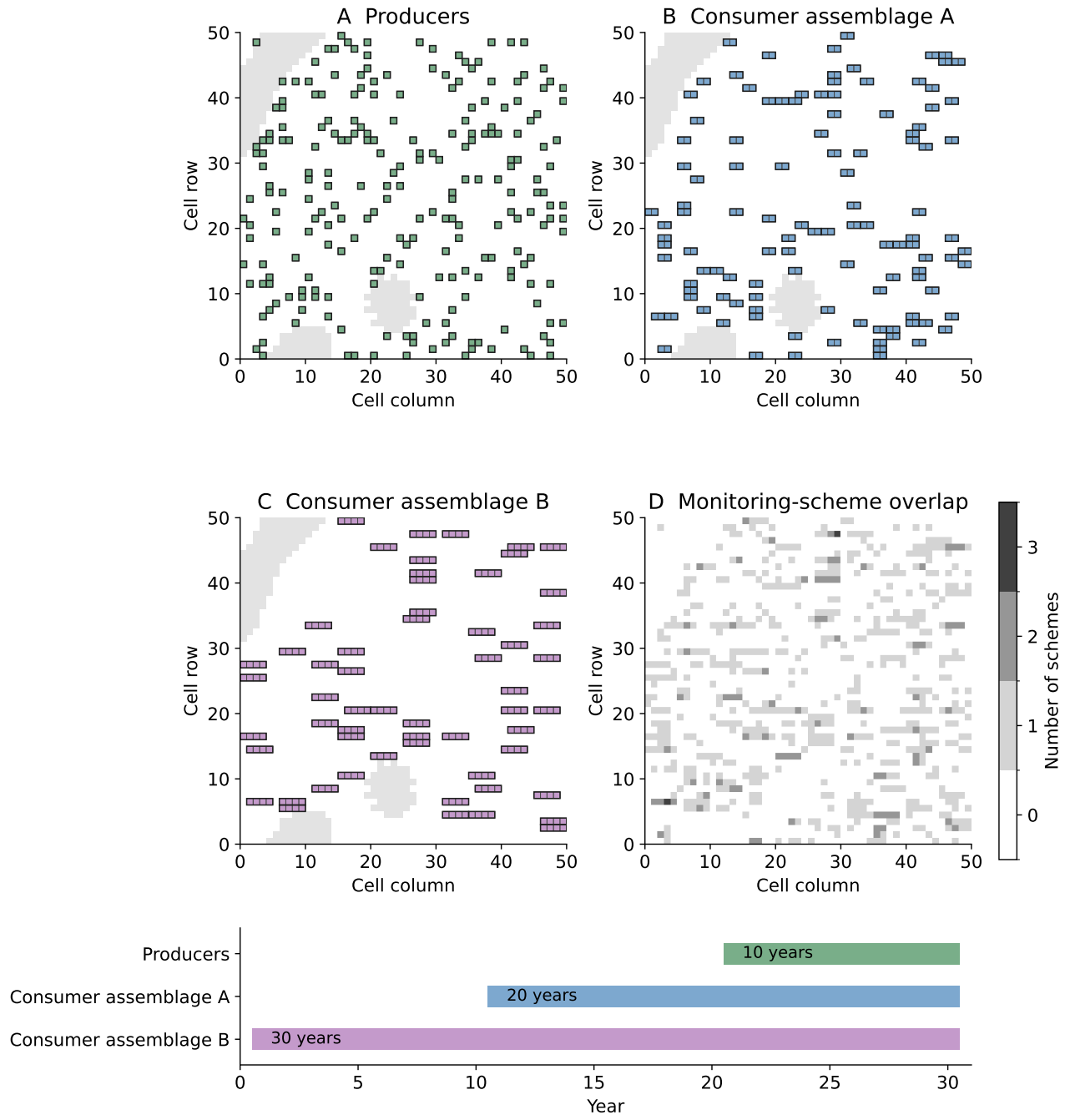


Figure 2: Monitoring footprints, scheme overlap and historical coverage. The four maps show producer monitoring (A), consumer assemblage A (B), consumer assemblage B (C) and the number of schemes covering each cell (D). In A–C, sampled cells have shaded centres and dark boundaries; grey cells were excluded by the ecological screen. In D, zero schemes is white and three schemes is dark grey, with intermediate greys for one and two schemes. The timeline gives the ten-, twenty- and thirty-year monitoring windows. Overlap counts spatial coverage across the complete monitoring design, rather than simultaneous observations in a particular year.

2.3 State-space neural ordinary differential equation

SS-NODE links a neural transition model to the count likelihood, separating measurement variation from ecological dynamics (Auger-Méthé et al., 2021). A latent trajectory is reconstructed for each cell, and its constituent species are aggregated over the survey footprint only when evaluating an available observation. Thus a producer survey and a larger consumer survey can constrain overlapping parts of the same community without requiring simultaneous, complete assemblage samples. Unobserved years remain part of the integrated trajectory and contribute no direct observation term. The continuous-time neural representation builds on neural differential-equation methods (Chen et al., 2018; Rubanova et al., 2019), but the present experiment tests unequal annual monitoring histories rather than irregular within-year sampling.

The fitted process receives phylogeny, environmental histories, producer/consumer roles, approximate feeding information and partially observed masses. It receives no generating equations, environmental optima, hidden biomass trajectories or true Jacobians. Twenty-one masses are observed; nine use one phylogenetic conditional-mean imputation, with producer and pooled-consumer means estimated from observed masses. The completed masses define an approximate feeding template $\widetilde{F}$.

Six learned species coordinates p_i are initialised from leading phylogenetic components and regularised using the phylogenetic precision matrix. For log biomass z_i, the model learns

$$\frac{dz_i}{dt} = g_\vartheta(E, p_i) + h_\vartheta(B_i, p_i) + r_i(B, p, \widetilde{F}) \sum_{j \neq i} A_{ij} B_j s_j(B, p, \widetilde{F}), \tag{4}$$

$$A_{ij} = \widetilde{F}_{ij}\,\mathrm{softplus}(u_{ij}) - \widetilde{F}_{ji}\,\mathrm{softplus}(v_{ij}) + \frac{0.05}{n}\tanh(w_{ij}). \tag{5}$$

A shared pair network maps both species' coordinates and both directed feeding preferences to the three coefficients. The feeding-informed terms represent gains and losses, while a weak residual permits effects outside the template. Diagonal pair coefficients are zero, and residuals on unsupported pairs receive an additional penalty. Thus the template informs interactions without imposing an absolute feeding-link mask. It is informative ancillary structure, not a network discovered without prior information.

Environmental responses use species coordinates, E and E^2. Producer intrinsic responses are unrestricted; consumer intrinsic responses are nonpositive. Learned own-density responses are negative for all species and can approach zero. This is a broad approximation prior, distinct from the generator's producer-only explicit density regulation. Receiver and sender gates range from 0.05 to 2 and use species coordinates, own log(1+B), available-resource biomass and consumer biomass. These gates allow local derivatives to change with community state. No additional saturation transformation of interaction biomass is imposed. The model uses single tanh hidden layers of 24 units for environment, 12 for density and 16 for pair effects and gates. It is integrated in log biomass using two exponential midpoint substeps per year.

2.4 Fitting and withheld-location assessment

The original random, interspersed spatial split assigns 68 of the 100 tiles to training, 16 to validation and 16 to testing. All monitoring schemes within a tile share its partition, and footprints remain within tiles, preventing sampled-cell overlap across partitions. The training and validation partitions contain 45,810 and 9,540 species-year counts, respectively; the test partition contains 11,880 counts. This evaluates reconstruction at withheld locations within the represented environmental domain rather than distant geographic extrapolation (Roberts et al., 2017).

We restore the saved, validation-selected checkpoint fitted under this split. Shared-parameter fitting uses the entire training dataset and uninterrupted trajectories through the common initial condition and ten-year burn-in. The mean Poisson loss is multiplied by 0.1 relative to regularisation. AdamW uses an initial learning rate of 0.002, weight decay 0.001 and gradient clipping at 5, with validation every ten epochs and plateau-based learning-rate reduction (Loshchilov and Hutter, 2019). Two fixed random starts receive up to 1,000 epochs each; validation likelihood selects the start and checkpoint. The present assessment reuses that checkpoint without further shared-parameter fitting.

Each test location is reconstructed twice. Observation-assisted reconstruction supplies all available monitoring fragments solely to infer local community states, keeping shared dynamics, species embeddings, interaction parameters and detection coefficients frozen. Ancillary-only reconstruction supplies the same environmental histories and species information but no local abundance observations. Thus observation availability varies between the two modes while the test locations, ecological truth and shared fitted model are identical.

Observation-assisted reconstruction uses regularised state-space smoothing over the full monitoring period. Let z^0_{ict} be the ancillary-only log-biomass trajectory. Smoothed states are $z_{ict} = z^0_{ict} + h_{k(c),i,t}$, where $k(c)$ identifies the 5 by 5 cell (1 km) tile. A species-specific correction is shared by all sampled cells in a tile at a given year, retaining within-tile environmental heterogeneity from the neural ODE. This prevents an aggregate observation from freely reallocating biomass among individual constituent cells.

The smoothing objective combines the footprint-aware Poisson likelihood with Gaussian penalties on departures from the frozen ODE transition. Transition residuals are $z_{t+1} - f_\vartheta(z_t, E_t)$, where f_ϑ is the annual transition in log biomass. Squared residuals are averaged across cells within each tile before summing, avoiding duplicate prior weight for tiles with more sampled cells. Additional Gaussian penalties anchor the initial correction with standard deviation 0.1 and all trajectory corrections with standard deviation 0.5 in log biomass. These discrepancy penalties accommodate approximation error in the fitted process; ecological process noise remains absent from the generating system. Adam uses 400 updates, learning rate 0.02 and gradient clipping at 10. Shared dynamics and detection parameters are frozen.

The transition standard deviation is selected from 0 (no correction), 0.01, 0.03 and 0.1 using masked validation counts only. One validation mask withholds an entire available monitoring assemblage per tile where other assemblages remain; another withholds a contiguous middle fifth of each monitoring history. Candidates are ranked by summed held-out Poisson deviance divided by the total number of held-out observations across the two masks. The chosen strength is then applied to all test fragments without further tuning. Neither test ecological truth nor interaction sensitivities enter selection or smoothing.

Both modes are assessed at the same 105 unique constituent cells of the 70 test footprints. For biomass and membership, both use a common mask excluding every species–cell–year entry that contributed to a count supplied in the observation-assisted mode. The exclusion also applies to ancillary-only reconstruction, even though that mode receives no counts, ensuring identical missing-information targets. Entries observed only through footprint averages are conservatively excluded as well. The common mask contains 79,650 assessed entries. Supplied-fragment fit is not reported as missing-information recovery.

Community richness counts all species above biomass 0.02 and uses identical complete cell-year communities in both modes. Membership F1, precision and recall use the common missing-information mask. Biomass correlations use log(1+biomass), with original-scale correlation and RMSE as complementary summaries. Annual correlations are calculated separately within each year, across the same

assessed entries in the two modes. Figures show consistent colours and partially transparent points with small horizontal offsets and jitter, without error bars; all scores use unjittered data.

Interactions are sensitivities of annual biomass change,

$$J_{ij}(B, E) = \frac{\partial\{\Phi_{1\,\mathrm{year},i}(B, E) - B_i\}}{\partial B_j}. \tag{6}$$

These include indirect within-year effects and differ from instantaneous feeding coefficients (Deyle et al., 2016). Both reconstruction modes assess the same off-diagonal entries at the original 64 prespecified test cells over 30 annual intervals. True Jacobians use true biomass; learned Jacobians use each mode's reconstructed biomass. Annual correlations pool identical site–species-pair entries within each interval. Strong-link sign accuracy uses true co-presence above biomass 0.02 and absolute true sensitivity above 0.01. Two example sites are selected from the 20th and 80th environmental percentiles independently of reconstruction quality and shown under both modes. Jacobians differentiate the frozen ecological transition, not the smoothing corrections or the posterior state estimator. No interaction targets enter fitting, checkpoint selection or smoothing.

3 Results

The original validation-selected model was restored, retaining seed 71622 at epoch 950. Masked validation selected a transition standard deviation of 0.01 for observation-assisted smoothing. Pooled held-out validation deviance per observation was 6.003, compared with 6.586 without corrections. Temporal validation deviance was 2.790 versus 6.400, while taxonomic validation deviance was 7.895 versus 6.696; selection therefore reflected a temporal improvement despite poorer taxonomic transfer. Both reconstruction modes assessed the same 105 test cells. The common missing-information mask excluded 18,000 entries contributing to supplied counts and retained 79,650 species–cell–year entries in each mode.

The correlation between true and reconstructed log(1+biomass) for missing entries was 0.959 with observation assistance and 0.959 with ancillary information alone (Figure 3). Original-scale correlations were 0.909 and 0.908, with RMSE 0.697 and 0.700, respectively. Annual estimates compare the same missing entries within each year.

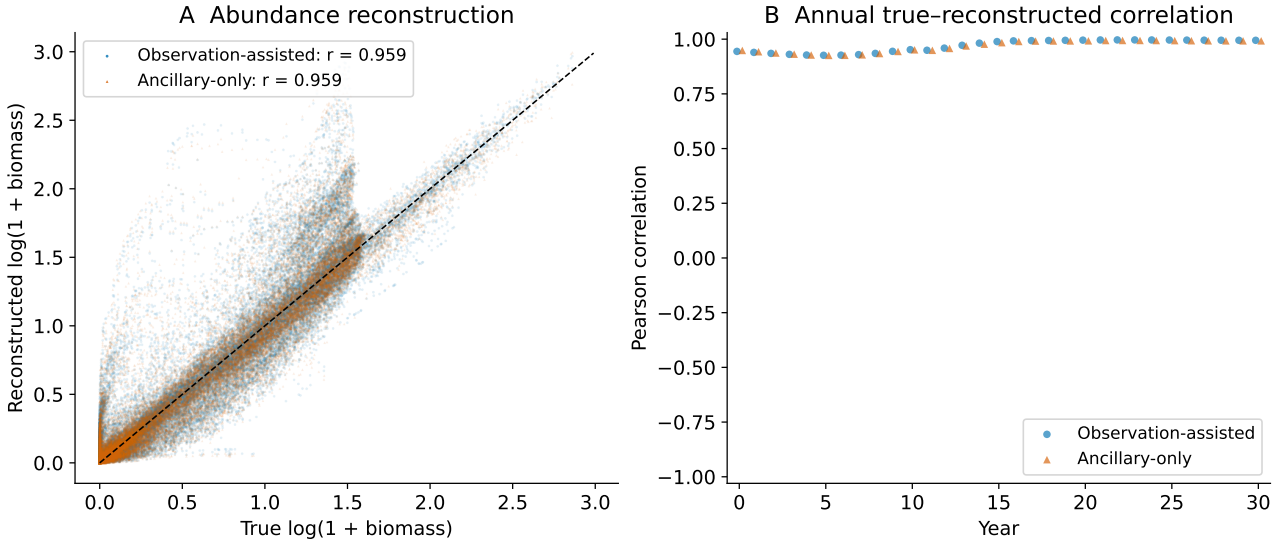


Figure 3: Reconstruction of missing biomass under two information settings. Blue circles identify observation-assisted reconstruction and orange triangles ancillary-only reconstruction, consistently throughout Figures 3–6. Both use the same test cells and the same species–cell–year scoring mask, excluding every entry that contributed to a monitoring count supplied to the observation-assisted mode. A, true versus reconstructed log(1+biomass), with separate Pearson correlations and transparent points. B, annual correlations, with small horizontal offsets and jitter. Correlations use unjittered values; no error bars are shown.

Full-community richness correlations were 0.976 and 0.973, with mean absolute errors of 1.42 and 1.50 species for observation-assisted and ancillary-only reconstruction, respectively (Figure 4). Richness includes all species at identical cell-years and complements the assessment of missing species identities.

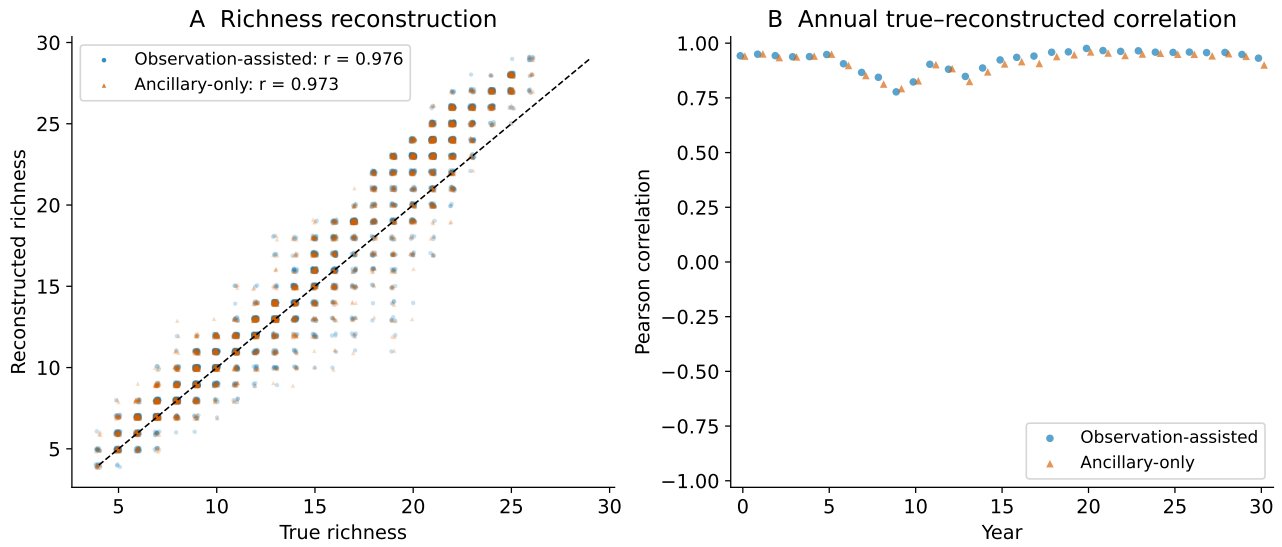


Figure 4: Full-community richness reconstruction at identical test locations. A, true versus reconstructed richness for observation-assisted (blue circles) and ancillary-only (orange triangles) reconstruction. Both use all species at the same cell-years. Small display jitter separates integer values; correlations use unjittered richness. B, annual correlations, with horizontally offset, jittered and partially transparent points. Membership requires biomass above 0.02.

Missing-membership F1 was 0.938 with observation assistance and 0.934 with ancillary information alone. Precision was 0.907 and 0.902, and recall was 0.973 and 0.970, respectively (Figure 5). All membership scores use the same exclusion mask.

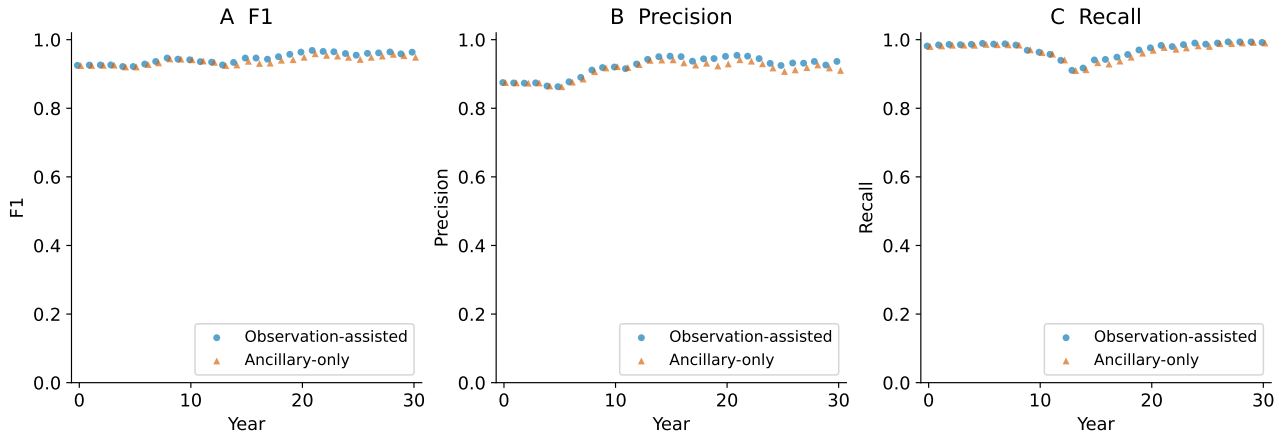


Figure 5: Annual recovery of missing species membership. Panels show F1, precision and recall for observation-assisted (blue circles) and ancillary-only (orange triangles) reconstruction. Both modes use an identical mask excluding entries that contributed to the supplied monitoring fragments. Points are horizontally offset, jittered and partially transparent, without error bars.

Correlations between true and reconstructed interaction sensitivities were 0.868 and 0.862 for observation-assisted and ancillary-only reconstruction, respectively (Figure 6). Strong-link sign agreement was 92.9% and 92.8%, while median reconstructed-to-true matrix norm ratios were 1.277

and 1.270. Both modes evaluated the same interaction entries, using their respective reconstructed community states.

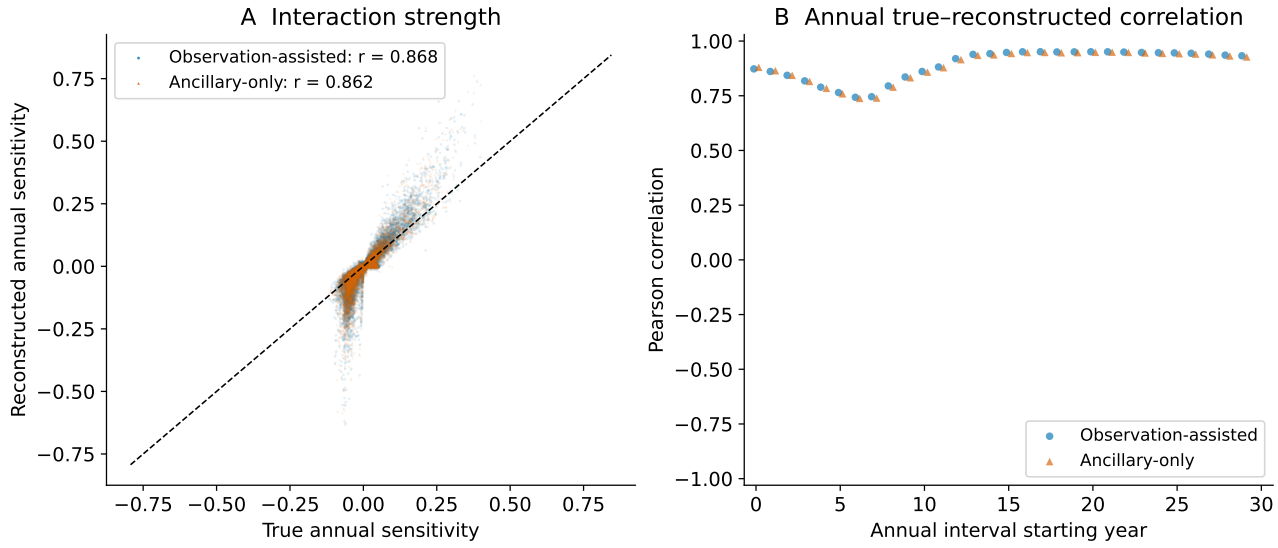


Figure 6: Recovery of local dynamical interaction sensitivities. A, true versus reconstructed off-diagonal annual-change Jacobians for observation-assisted (blue circles) and ancillary-only (orange triangles) reconstruction. Both modes assess identical species-pair entries at the same 64 cells and 30 annual intervals. Up to 70,000 entries per mode are plotted with transparency; correlations use all entries. B, annual correlations with horizontal offsets and jitter. True Jacobians use true community states, whereas learned Jacobians use reconstructed states. Jacobians are derivatives of the frozen ecological transition at reconstructed states, excluding smoothing corrections. No Jacobians inform fitting or smoothing.

Figure 7 compares the two modes at the same two example sites at year 15. The repeated true matrices provide a common reference for the reconstructed matrices and their signed errors.

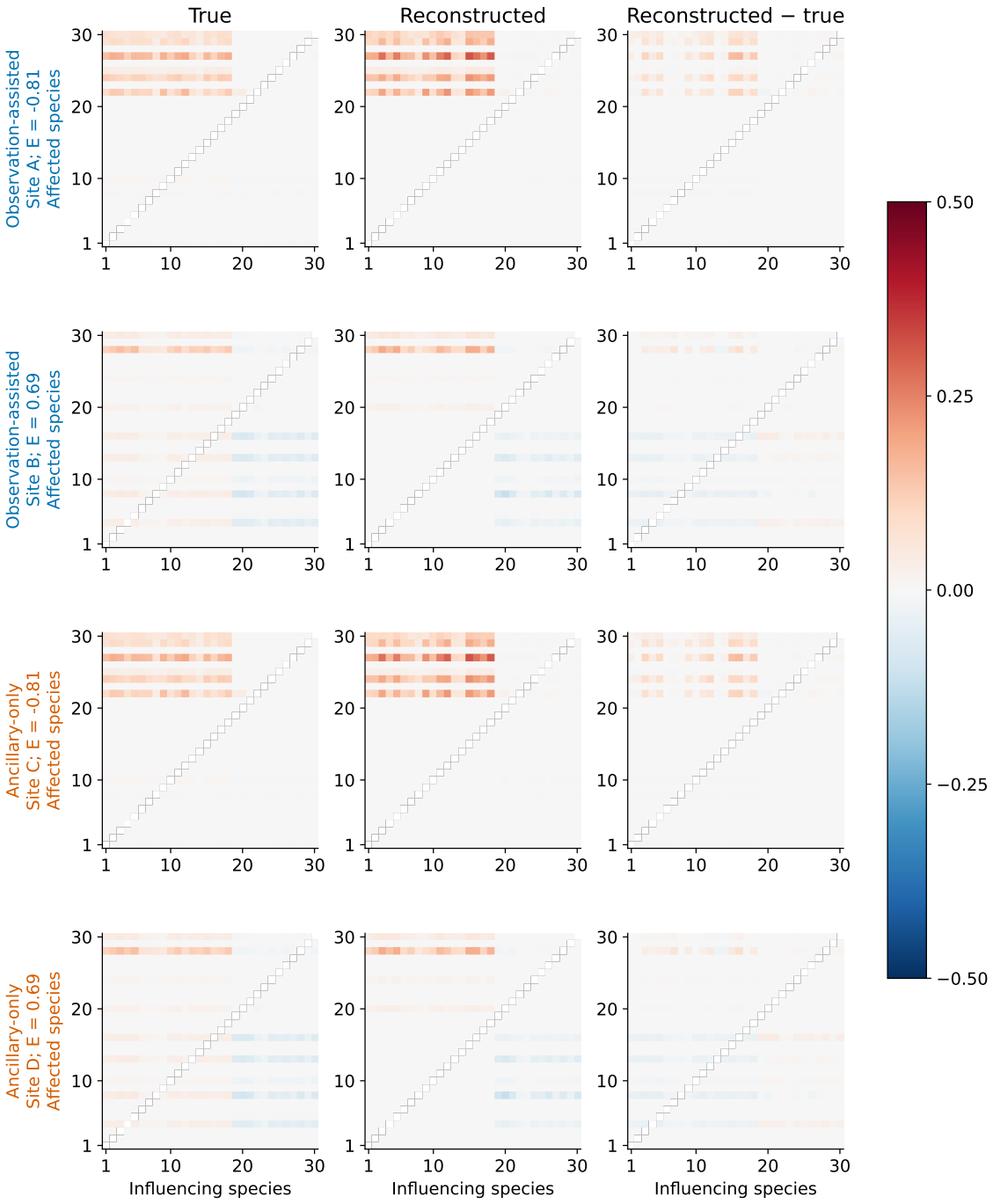


Figure 7: Local interaction matrices under observation-assisted and ancillary-only reconstruction. The first two rows show observation-assisted reconstruction at two test sites; the final two show ancillary-only reconstruction at the same two sites in the same order. Sites are selected from environmental quantiles independently of reconstruction quality. At year 15, columns show true annual-change Jacobians, reconstructed Jacobians and reconstructed minus true values. One shared, zero-centred colour bar is fixed at -0.5 to 0.5, with larger magnitudes saturating at the endpoints. Diagonals are masked. Rows are affected species and columns influencing species.

4 Discussion

SS-NODE combines an explicit observation model with a shared dynamical approximation to reconstruct communities from partial monitoring records. The matched assessment distinguishes the contribution of local observations from transfer using ancillary information: both modes use the same test cells, missing-information mask and ecological model. Observation-assisted smoothing achieved missing-biomass correlation 0.959 and interaction correlation 0.868, compared with 0.959 and 0.862 under ancillary-only reconstruction. These scores describe the present controlled simulation and inference procedure, rather than a general guarantee that conditioning improves hidden-state recovery.

State-space smoothing allows observations to constrain communities near their observation times, with deviations from the approximate process penalised throughout the trajectory (Auger-Méthé et al., 2021). Spatial pooling retains the observation operator's footprint averages without giving each constituent cell an independent correction. However, these restrictions do not make missing species uniquely identifiable. Observations of one assemblage can constrain another through the learned dynamics only to the extent that the relevant relationships have been recovered. Validation selection includes the unmodified trajectory as a safeguard against corrections that do not generalise to masked observations.

The interaction matrices remain derivatives of one frozen ecological model, evaluated at the reconstructed states. They therefore vary with environment and composition, without a separate interaction matrix being fitted at each location (Poisot et al., 2015; Deyle et al., 2016). The smoothing corrections are excluded from these derivatives. Better state estimation can improve the states at which sensitivities are evaluated, but cannot repair an incorrect shared interaction function. These annual-change sensitivities include indirect effects and do not establish uniquely identified causal interactions (Blanchet et al., 2020).

The experiment supplies known environmental histories, observation footprints, count conversion and informative species structure. Detection follows the fitted family, ecological process noise is absent and monitoring is restricted to persistent multi-assemblage communities. Single body-mass imputations and point estimates of smoothed states do not propagate uncertainty. Gaussian discrepancy penalties in inference accommodate errors of approximation rather than demonstrating stochastic ecological dynamics. Spatial grouping prevents sampled-cell overlap across partitions, but the interspersed split evaluates reconstruction within the represented environmental domain (Roberts et al., 2017). Broader simulations and independent monitoring data are needed to assess generality.

Shared parameters avoid an independent ecological fit at every location, while the pooled smoother estimates local states under those shared dynamics. Dense pairwise ecological interactions remain quadratic in species number, and large-data throughput has not been tested.

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